

WARNING: MISBEHAVIOR AT EXAM TIME WILL LEAD TO SERIOUS CONSEQUENCE.

SCUT Final Exam

20241212 《Calculus I》 REVIEW Exam Paper B

- Notice:
1. Make sure that you have filled the form on the left side of seal line.
 2. Write your answers on the exam paper.
 3. This is a close-book exam.
 4. The exam with full score of 100 points lasts 120 minutes.

Question No.	1	2	3	4	5	6	7	8	9	Sum
Score										

1. Evaluate the following limits.

- (1) $\lim_{x \rightarrow \infty} \left(\frac{x^2+1}{x^2-2} \right)^{x^2};$
- (2) $\lim_{x \rightarrow 0} \frac{1-x^2-e^{-x^2}}{x \sin^3 x};$
- (3) $\lim_{x \rightarrow 8} \frac{\sqrt{9+2x}-5}{\sqrt[3]{x}-2};$
- (4) $\lim_{n \rightarrow +\infty} \left(\sin \frac{\pi}{\sqrt{n^2+1}} + \sin \frac{\pi}{\sqrt{n^2+2}} + \cdots + \sin \frac{\pi}{\sqrt{n^2+n}} \right).$

2. Evaluate the problems.

(1) Let $f(x) = x^2 \sin x$, find $f^{(5)}(0)$.

(2) Let $f(x)$ be differentiable at x_0 , please find

$$\lim_{x \rightarrow 0} \frac{f(x_0 + x) - f(x_0 - x)}{x}.$$

(3) If the parametric equation $\begin{cases} x = a \cos^3 t \\ y = a \sin^3 t \end{cases}$ defines y as a differentiable function of x , find $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$.

(4) If $\int_0^y e^{t^2} dt + \int_0^{\sin x} \cos^2 t dt = 0$, find dy .

3. Evaluate the following integrals.

(1) $\int \sin(\ln x) dx$;

(2) $\int \frac{1}{x^4 \sqrt{1+x^2}} dx$;

(3) $\int_{\frac{1}{3}}^{\frac{2}{3}} e^{\sqrt{3x-1}} dx$,

(4) $\int_{-1}^1 \frac{x^2(1+\sin x)}{1+\sqrt{1-x^2}} dx$;

(5) $\int_0^{+\infty} \frac{dx}{(x+1)(x+2)}$.

4. Try to prove that $\lim_{x \rightarrow a} x^3 = a^3$ by using $\varepsilon - \delta$ definition.

5. Let $f(x)$ be integrable on $[a, b]$, and $\forall x_1, x_2 \in [a, b]$, we have

$$|f(x_1) - f(x_2)| \leq |x_1 - x_2|,$$

try to prove that

$$\int_a^b f(x) dx \leq (b-a)f(a) + \frac{1}{2}(b-a)^2.$$

6. If $f(x), g(x) \in C[a, b]$, and $g(x) \geq 0$, try to prove that $\exists c \in$

$[a, b]$, such that $\int_a^b f(x) g(x) dx = f(c) \int_a^b g(x) dx$.

7. Let $f(x) \in C[a, b]$, $f(x)$ be differentiable on (a, b) . $f(a) = f(b)$, $f(x)$ is not a constant function. Try to prove that $\exists c \in (a, b)$, s.t. $f'(c) < 0$.

8. Find the area of the surface generated by revolving the given curve $x = t, y = t^3, (0 \leq t \leq 1)$ about the x -axis.

9. Let R be the region bounded by $y = x^2, y = x$. Find the volume of the solid that results when R is revolved around:

(1) The x -axis, (2) the y -axis.